Spring 1999

Problem 1 (Sp99). Let X and Y be nonempty subsets of a metric space M and $d(X, Y)$ the distance between X and Y as defined in **??**. Give a proof or counterexample for each of the following statements, for disjoint sets X and Y.

1. If X and Y are closed in M then $d(X, Y) > 0$.

2. If X and Y are compact then $d(X, Y) > 0$.

3. If X is closed and Y compact then $d(X, Y) > 0$.

Problem 2 (Sp99). Suppose that a sequence of functions $f_n : \mathbb{R} \to \mathbb{R}$ converges uniformly on $\mathbb{R}$ to a function $f : \mathbb{R} \to \mathbb{R}$, and that $c_n = \lim\limits_{x \to \infty} f_n(x)$ exists for each positive integer n. Prove that $\lim\limits_{n \to \infty} c_n$ and $\lim\limits_{x \to \infty} f(x)$ both exist and are equal.

Problem 3 (Sp99). Suppose that f is a twice differentiable real valued function on $\mathbb{R}$ such that $f(0) = 0$, $f'(0) > 0$, and $f''(x) \geqslant f(x)$ for all $x \geqslant 0$. Prove that $f(x) > 0$ for all $x > 0$.

Problem 4 (Fa79, Fa80, Sp85, Su85, Fa96, Fa98, Sp99, Sp26). Prove that

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x}\,\mathrm{d}x = \frac{\pi}{\sin \pi\alpha}$$

What restrictions must be placed on α?

Problem 5 (Sp99, Fa11).

1. Prove that if f is holomorphic on a simply connected region U, and $f(z) \neq 0$ for all $z \in U$, then there is a holomorphic function g on U such that $f(z) = e^{g(z)}$ for all $z \in U$.

2. Does the conclusion of Part 1 remain true if U is replaced by an arbitrary connected open subset of $\mathbb{C}$?

Problem 6 (Sp99). Let p, q, r and s be polynomials of degree at most 3. Which, if any, of the following two conditions is sufficient for the conclusion that the polynomials are linearly dependent?

1. At 1 each of the polynomials has the value 0.

2. At 0 each of the polynomials has the value 1.

Problem 7 (Sp99). Suppose that the minimal polynomial of a linear operator T on a seven-dimensional vector space is x^2. What are the possible values of the dimension of the kernel of T?

Problem 8 (Sp99). Let M be a 3×3 matrix with entries in the polynomial ring $\mathbb{R}[t]$ such that $M^3 = \begin{pmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix}$. Let N be the matrix with real entries obtained by substituting $t = 0$ in M. Prove that N is similar to $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

Problem 9 (Sp99). Let G be a finite group, with identity e. Suppose that for every $a, b \in G$ distinct from e, there is an automorphism σ of G such that $\sigma(a) = b$. Prove that G is abelian.

Problem 10 (Sp99). Suppose that f is a twice differentiable real function such that $f''(x) > 0$ for all $x \in [a, b]$. Find all numbers $c \in [a, b]$ at which the area between the graph $y = f(x)$, the tangent to the graph at $(c, f(c))$, and the lines $x = a$, $x = b$, attains its minimum value.

Problem 11 (Sp99). Prove that if n is a positive integer and α, ε are real numbers with $\varepsilon > 0$, then there is a real function f with derivatives of all orders such that

- $|f^{(k)}(x)| \leqslant \varepsilon$ for $k = 0, 1, \ldots, n-1$ and all $x \in \mathbb{R}$,
- $f^{(k)}(0) = 0$ for $k = 0, 1, \ldots, n-1$,
- $f^{(n)}(0) = \alpha$.

Problem 12 (Sp99). Suppose that f is holomorphic on some neighborhood of a in the complex plane. Prove that either f is constant on some neighborhood of a, or there exist an integer $n > 0$ and real numbers $\delta, \varepsilon > 0$ such that for each complex number b satisfying $0 < |b - f(a)| < \varepsilon$, the equation $f(z) = b$ has exactly n roots in $\{z \in \mathbb{C} \mid |z - a| < \delta\}$.

Problem 13 (Sp99). Let $b_1, b_2, \ldots$ be a sequence of real numbers such that $b_k \geqslant b_{k+1}$ for all k and $\lim_{k\to\infty} b_k = 0$. Prove that the power series $\sum_{k=1}^{\infty} b_k z^k$ converges for all complex numbers z such that $|z| \leqslant 1$ and $z \neq 1$.

Problem 14 (Sp99). Let $A = \left(a_{ij}\right)$ be a $n \times n$ complex matrix such that $a_{ij} \neq 0$ when $i = j + 1$, $a_{ij} = 0$ when $i \geqslant j + 2$. Prove that A cannot have more than one Jordan block for any eigenvalue.

Problem 15 (Sp99). Let M be a square complex matrix, and let $S = \{XMX^{-1} \mid X \text{ is nonsingular}\}$ be the set of all matrices similar to M. Show that M is a nonzero multiple of the identity matrix if and only if no matrix in S has a zero anywhere on its diagonal.

Problem 16 (Sp99). Let $\|x\|$ denote the Euclidean norm of a vector x. Show that for any real $m \times n$ matrix M there is a unique nonnegative scalar σ, and (possibly nonunique) unit vectors $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^m$ such that

- $\|Mx\| \leqslant \sigma \|x\|$ for all $x \in \mathbb{R}^n$,
- $Mu = \sigma v$,
- $M^\top v = \sigma u$.

Problem 17 (Sp99). Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of degree $n \geqslant 3$. Let $\mathbb{L}$ be the splitting field of f, and let $\alpha \in \mathbb{L}$ be a zero of f. Given that $[\mathbb{L} : \mathbb{Q}] = n!$, prove that $\mathbb{Q}\left(\alpha^4\right) = \mathbb{Q}(\alpha)$.

Problem 18 (Sp99). Let G be a finite simple group of order n. Determine the number of normal subgroups in the direct product $G \times G$.